

Omar Alberto Murillo Moreno

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SUMMARY

Full-stack engineer specializing in modern web applications, APIs, and AI agent platforms. Experienced in developing and deploying **Microsoft Copilot Agents**, **MCP servers**, and scalable backend services using **React**, **TypeScript**, **JavaScript**, **Go**, and **SQL**. Strong background in cloud infrastructure, Docker, AWS, and Kubernetes, with a focus on building reliable, maintainable, and high-performance systems.

EDUCATION The technological institute of Ciudad Juarez Juarez City, Chihuahua/Mexico

Bachelor of Mechatronic Engineering

05-2024

- **GPA:** 9.72
- **Relevant Coursework:** Mechanical Design (SolidWorks), Deep Learning course provided by Microsoft

WORK & LEADERSHIP EXPERIENCE Robert Bosch Juarez, Chihuahua/Mexico

Software developer Technician, 09-2023 – Present

- Led the design and deployment of **Microsoft Copilot Agents**, incorporating knowledge grounding, tool integrations, and instruction frameworks to provide intelligent, context-aware assistance. Architected **MCP services** in **TypeScript** and **Python** connected through a **Go-based API Gateway**, accelerating data access workflows and reducing information retrieval times from minutes to seconds.
- Supported third-party integrations by optimizing API calls, reducing time access the proving documentation and maintainability, including troubleshooting and consuming for RabbitMQ services.
- Designed a highly scalable API Gateway in Go, following SOLID principles and leveraging built-in features to enhance performance and reduce latency.
- **React Modernization Project:** Refactored and restructured legacy React applications using SOLID principles, modular architecture, and best practices for styling and TypeScript/JavaScript environments.
- **Production Middleware Integration:** Reverse-engineered sealed production software to understand communication protocols. Developed WCF SOAP services on IIS as middleware between .NET Core APIs and third-party systems, integrating seamlessly with SAP.
- **Laboratory Data Platform:**
 - ❖ Designed Oracle database schema, triggers, and procedures per company standards.
 - ❖ Implemented Express-based backend using Clean Architecture for improved scalability and database flexibility.
 - ❖ Deployed backend, frontend, and FTP services using Docker and Kubernetes.
 - ❖ Enhanced Angular modules and directives for intuitive, user-centered UI.

- ❖ Troubleshoot Nginx and Docker issues on RHEL servers to ensure stable deployments.

Key Technologies: React, JavaScript, TypeScript, Node.js, Express.js, .NET Core, Oracle SQL, Docker, Kubernetes, SAP Integration, WCF, Angular, Nginx, Microsoft Copilot, Power Automate, MCP, Python

Inteligene **Juarez, Chihuahua/Mexico** *Junior*
developer 05-2021 – 09-2023

- Implemented OCR capabilities using AI research and Tesseract for improved text recognition performance.
- Created and maintained technical documentation for manual and automated workflows.
- Designed SQL-optimized queries and integrated data visualizations using D3.js and other JavaScript libraries.
- Developed APIs with Fastify and Express, leveraging Sequelize ORM for database management.
 - Integrated real-time communication with Socket.IO and front-end modules using React.
- Built and deployed Flask-based microservices on AWS Lambda via Docker, ECR, and automated pipelines (CodeBuild, CodeDeploy, CodeCommit).

Key Technologies: React, SQL, Fastify, Express, Sequelize, Flask, Docker, AWS Lambda, ECR, D3.js, Tesseract.js, Socket.IO.

Web developer assistant 03-2020 – 05-2022

- Developed and maintained manufacturing websites ensuring compliance with Mexican regulations.
- Deployed Ruby on Rails applications on AWS Elastic Beanstalk with CI/CD pipelines.
- Migrated R-based applications to AWS Lambda microservices, cutting monthly costs by 50%.

ASME **Juarez, Chihuahua/Mexico**

Assistant, Student 04-2018 – 01-2019

- Provided assistance in college-fund events by organizing and setting up the event.
- Programmed robot movements on Arduino IDE as well as helped build its exoskeleton.

SKILLS, ACTIVITIES & INTERESTS

Languages: Fluent in Spanish; Conversational Proficiency in English **Technical**

Skills:

- **Frontend:** React, Angular, Vue, HTML5, CSS3, JavaScript (ES6+), TypeScript
- **Backend:** Node.js (Express, Fastify), .NET Core / Framework (C#), Flask, Ruby on Rails, Python (FastAPI)
- **Databases:** MySQL, SQL Server, Oracle, MongoDB
- **DevOps & Cloud:** Docker, Kubernetes (basic), AWS (Lambda, ECS, ECR, Elastic Beanstalk, S3, Textract), Microsoft Copilot, Power automate, MCP Protocol.
- **Other Tools:** Git, Nginx, REST/SOAP APIs, WebSockets

Interests: Continuous learning in software architecture, design patterns, and DevOps. Enjoys creative work such as clay modeling, diorama making, and painting.